

Guided Lab 3.1 — Clustering and Pattern Discovery in Cybersecurity Data

Course 02 – Introduction to Machine Learning for Cybersecurity

Lab Goal

In this guided lab, students will explore unsupervised learning techniques used to identify hidden patterns and anomalies in cybersecurity-related datasets. Students will apply clustering algorithms such as K-Means and DBSCAN to group similar records and investigate unusual behavior.

The lab also introduces dimensionality reduction using PCA to help visualize high-dimensional cybersecurity data.

Learning Objectives

By completing this lab, students will be able to:

- Understand the purpose of unsupervised learning
- Apply K-Means clustering to cybersecurity data
- Explore DBSCAN clustering for anomaly detection
- Use PCA for dimensionality reduction and visualization
- Interpret clusters and unusual patterns in a cybersecurity context
- Visualize machine learning outputs using Python

Technologies and Tools

- Python
- Jupyter Notebook / Google Colab
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Tips for Success

- Focus on understanding the purpose of clustering rather than memorizing syntax
- Carefully observe visualizations and patterns
- Compare how K-Means and DBSCAN behave differently
- Use comments in your notebook to explain your workflow
- Connect clustering results to cybersecurity scenarios

Dataset Overview

Students will work with a cybersecurity-related dataset containing login activity information.

[cybersecurity_login_dataset.csv](#)

Example Features:

- Login hour
- Failed login attempts
- Session duration
- IP reputation score
- New device indicator
- Unusual location indicator

The dataset does not contain labels for suspicious activity because this is an unsupervised learning activity.

Introduction to Clustering

Clustering is an unsupervised learning method that groups similar records together based on shared characteristics.

In cybersecurity, clustering can help:

- Identify suspicious login behavior
- Detect unusual network traffic patterns
- Group similar user activity
- Discover hidden attack patterns



K-Means Clustering

K-Means clustering groups data into a predefined number of clusters.

Example:

A login activity dataset may naturally separate into:

- Normal employee behavior
- High-frequency login attempts
- Unusual access patterns



DBSCAN Clustering

DBSCAN identifies clusters based on data density and can automatically detect outliers.

Why This Matters in Cybersecurity:

Outliers may represent:

- Suspicious logins
- Automated bot activity
- Potential intrusions
- Unusual network behavior



PCA (Principal Component Analysis)

PCA helps reduce complex datasets into fewer dimensions while preserving important information.

Benefits:

- Easier visualization
- Faster analysis
- Better understanding of patterns

✂ Guided Lab 3.1 Activities

Part 1 — Import Libraries

Students will:

- Import required Python libraries
- Set up the notebook environment

Part 2 — Load and Explore the Dataset

Students will:

- Load the dataset into a Pandas DataFrame
- Explore rows, columns, and summary statistics
- Check for missing values

Part 3 — Prepare the Dataset

Students will:

- Select numerical features
- Normalize or scale the data if necessary

Part 4 — Apply K-Means Clustering

Students will:

- Train a K-Means clustering model
- Create clusters from login activity data
- Visualize clusters using scatter plots

Discussion Questions:

- What patterns appear in the clusters?
- Which groups may represent unusual behavior?

Part 5 — Apply DBSCAN Clustering

Students will:

- Apply DBSCAN clustering
- Identify outliers and anomalies
- Compare DBSCAN results with K-Means

Discussion Questions:

- Which records appear isolated?
- Why might these points be suspicious?

Part 6 — Dimensionality Reduction with PCA

Students will:

- Apply PCA to reduce dataset dimensions
- Visualize the dataset in 2D space
- Observe how clusters and anomalies appear visually

Discussion Questions:

- Did PCA make patterns easier to understand?
- What cybersecurity insights could visualization provide?



Example Cybersecurity Interpretation

Suppose a cluster contains users who:

- Login late at night
- Use new devices
- Have repeated failed login attempts

This cluster may represent:

- Suspicious user behavior
- Compromised accounts

- Possible automated attacks

Another isolated point identified by DBSCAN may represent a rare or highly unusual login event worth investigating.



Submission Requirements

Submit:

- ✓ Completed Jupyter Notebook (.ipynb)
- ✓ All notebook cells executed with outputs visible
- ✓ Visualizations generated during the lab
- ✓ Short written interpretation answering:
 - What patterns did you observe?
 - Which records or clusters appeared suspicious?
 - How can unsupervised learning support cybersecurity monitoring?



Deliverable

- ✓ Guided Lab 3.1 Notebook Submission



Additional Learning Resources



Documentation

- Scikit-learn Clustering Documentation
<https://scikit-learn.org/stable/modules/clustering.html>
- PCA Documentation
<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>
- DBSCAN Documentation
<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.DBSCAN.html>



Beginner-Friendly Tutorials

- Google Machine Learning Crash Course
<https://developers.google.com/machine-learning/crash-course>
- Kaggle Intro to Machine Learning
<https://www.kaggle.com/learn>



Reflection Questions

1. What is the purpose of clustering in machine learning?
2. How does DBSCAN differ from K-Means?
3. Why is anomaly detection useful in cybersecurity?
4. How can PCA help analysts understand complex datasets?
5. What challenges might occur when using unsupervised learning in real-world cybersecurity systems?